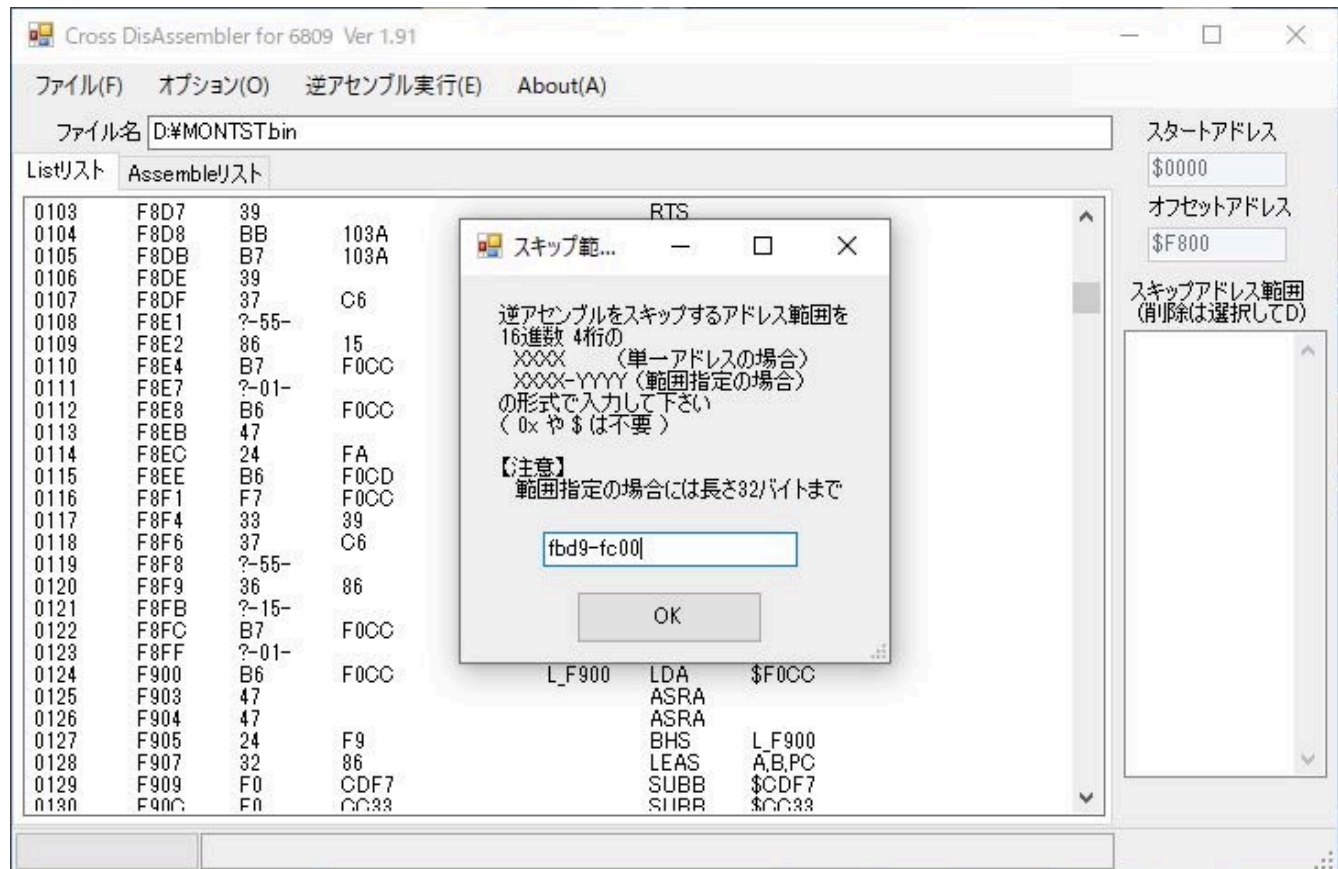


A revised version of the 6809/6800 cross-disassembler introduced earlier.

 flexonsbd.blogspot.com/2021/10/68096800.html

Saturday, October 2, 2021



A revised version of the 6809/6800 cross-disassembler that I introduced.

I've corrected the cross-disassembler for 6809/6800.

In my September 1st blog post, "Introducing the assembler and other tool software I created out of necessity," I mentioned a cross-disassembler, one of the pieces of software I created myself, but it has a bug.

In versions prior to the one released, the portion specified in the skip range was replaced byte by byte with an FCB pseudo-instruction, so in display messages and other contexts, this had to be manually replaced with an FCC pseudo-instruction.

For example, when the skip range data was specified as 36, 38, 30, 39, 53, 59, 53, 54, 45, 4D or 0D, 0A, 04

, it was written as FCB \$36 FCB \$0D FCB \$38 FCB \$0A FCB \$30 FCB \$04 FCB

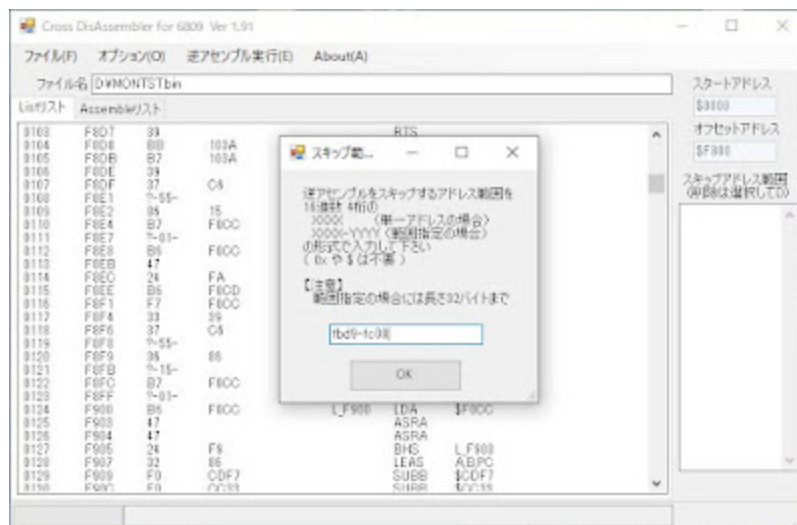
\$39 FCB \$53 FCB \$59 FCB \$53 FCB \$54 FCB \$45 FCB \$4D . However, to make the list easier to read, it was manually rewritten as FCC /6809SYSTEM/ or FCB \$0D, \$0A, \$04 .

To enable the disassembler itself to handle this, the released version replaces the first value of the data specified in the skip range with an FCC pseudo-instruction if it is displayable data between \$20 and \$7F, and with an FCB pseudo-instruction otherwise, and also allows multiple byte sequences to be combined into a single pseudo-instruction.

I rewrote the program to allow this substitution, but I forgot that the operand length of the resulting FCC and FCB pseudo-instructions is undefined and can be quite long in some cases. In particular, the FCB pseudo-instruction requires 4 bytes for every byte of data.

Therefore, in the released version, if the skip range was set too large, it exceeded the size of the string variable used to store the operands, causing the program to crash. (I didn't notice this during testing because I didn't specify such a large range... Incidentally, it's possible to specify up to 100 skip ranges, so this bug can be avoided by dividing them into smaller groups.)

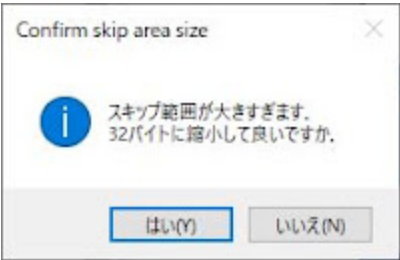
In this revised version, we've added a feature that checks the length of the skip range input and limits it to 32 bytes if it exceeds 32 bytes.



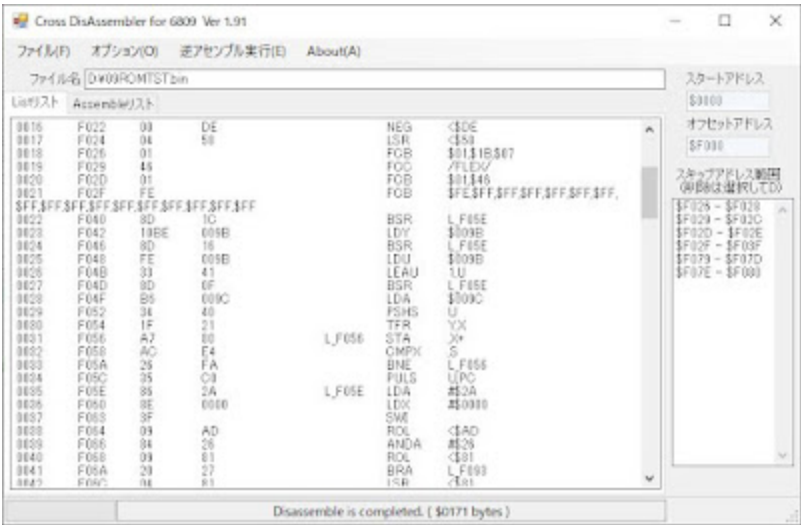
If the specified range exceeds 32 bytes,

This demonstrates the operation. The operands for the FCC and FCB pseudo-instructions consist of multiple bytes.

First, here's the 6809 version.

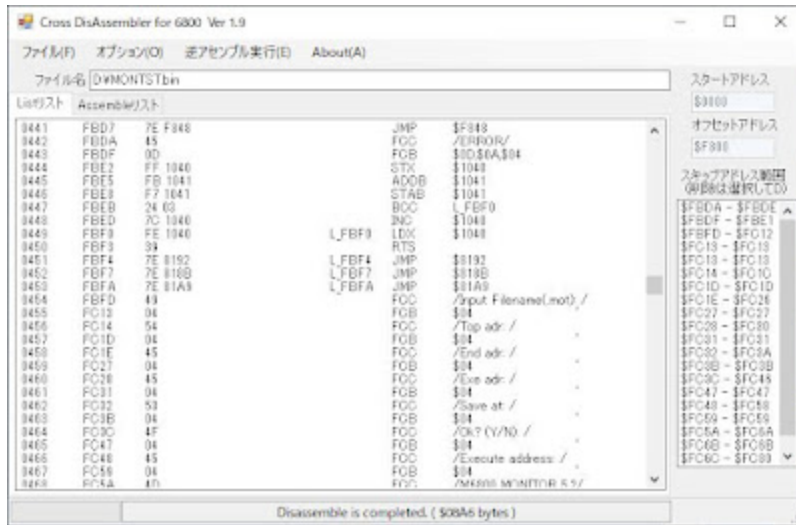


It will be reduced to 32 bytes.



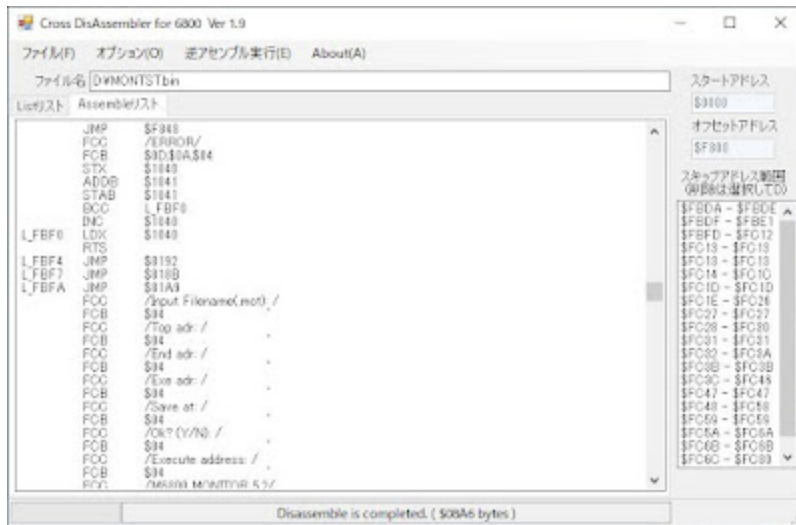
Lst version 6809

The same applies to the 6800 version.



Lst 6800 version

The Asm listing can be directly input into the assembler. The image shows an example for the 6800 version, but the 6809 version is similar.



6800 version Asm list

I'll upload the corrected version to [OneDrive](#).